CMPS 360 - Data Science Fundamentals

Assignment 1: Hands-on Data Engineering

Individual assignment | 100 marks – **Deadline 12noon Thursday 8 October**

In this assignment, you will act as a junior data engineer tasked to build a small, reproducible pipeline to create a Lakehouse using DuckLake or Databricks. You will choose a public dataset, define four questions, and implement a Bronze-to-Silver-to-Gold pipeline. The final Gold layer must use a star schema and support the four questions.

1. Dataset Selection

Select a dataset from a reliable, publicly accessible source. Choose a topic that interests you such as transport, environment, public services, sports, energy, education, or tourism. Avoid retail if similar to the example we did together in class.

Suggested source	Examples
Government open-data portals	~~Qatar Open Data,~~ data.gov, data.gov.uk, other official national or municipal portals
International organizations	World Bank Open Data, World Health Organization, United Nations data portals
Academic repositories	UCI Machine Learning Repository
Curated public repositories	Kaggle Datasets

Discuss and get approval of your selected dataset from the instructor. Max 3 students may work on the same dataset but focus on answering different questions.

2. Four Analytical Questions

Define four concrete, meaningful, and answerable questions before building the pipeline. These questions guide your Silver transformations, Gold model, and SQL queries.

- Each question must be answerable from the selected dataset after reasonable cleaning.
- Collectively, the questions should use at least two dimensions, such as ~~time~~date, category, location, product, customer, or service type.
- At least one question must involve a comparison, at least one must involve ordered ranking when the data supports it, and at least one must use an aggregate measure such as count, sum, average, minimum, or maximum.
- Do not ask prediction, classification, clustering, causal-inference, or advanced-machine-learning questions.

Example forms How did X change over time? Which categories contribute most to Y? How does Y differ by region? Which entities have the highest or lowest average Z? Adapt these forms to your dataset.

3. Assignment Tasks

Task 1 - Understand the source and define the grain (8 marks)

1. In a README.md file or ~~at the beginning of your notebook,~~ include the dataset title, source organization, direct URL, ~~access date,~~ number of files, file format(s), number of rows and columns, and a short explanation of why the dataset is interesting.
2. Describe what one source row represents. This is the source grain.

3. Identify the candidate key or combination of columns that should uniquely identify a row.
4. List the important columns needed to answer your questions, expected data types, units, and meanings in a compact data dictionary.

Task 2 - Design the ingestion strategy (12 marks)

Explain the most appropriate design for your dataset. Address all three decisions below, even if you implement only a local batch pipeline.

Decision	Choose and justify
Full vs. incremental	Choose full, incremental, or full first then incremental. Identify a possible watermark for incremental loading, if applicable.
Batch vs. streaming	Streaming may be discussed if applicable, but implementation is not required.
Push vs. pull	Decide and justify who initiates delivery.

- State a reasonable ingestion frequency, such as weekly, daily, or on demand.
- You may simulate a second batch by splitting the dataset into files for different date ranges. This is recommended when demonstrating an incremental design.

Task 3 - Build the Bronze layer (10 marks)

- Add tracking metadata such as source_name, ingested_at, and load_id.
- For each source file, record the input row count for each load.

Task 4 - Build the Silver layer (20 marks)

Create trusted, analysis-ready tables from Bronze. Apply only transformations that are relevant to your dataset and explain each decision. Quarantine invalid rows. Create useful derived columns only when they support the analytical questions. Apply at least three meaningful validation rules.

Required quality evidence: Include a compact quality report showing, at minimum: Bronze row count, Silver valid row count, quarantined row count, duplicate count, null counts for critical columns, and uniqueness of the selected key.

Task 5 - Design and implement the Gold star schema (20 marks)

Create a simple star schema from Silver. The Gold layer must be designed around your four analytical questions. Define the fact-table and its grain. Create appropriate 3 to 4 dimension tables that describe the facts. Provide a readable star-schema diagram.

Keep the star simple: Do not create dimensions that do not support the four questions. Do not copy the full Silver table into every Gold table. Prefer one clear fact table connected directly to descriptive dimensions.

Task 6 - Answer the four questions (20 marks)

Use only Gold-layer tables to answer the four analytical questions. Write one clearly labeled SQL query for each question. For every question, write a short evidence-based interpretation of the result.

Task 7 - Reflect on the design (5 marks)

Write 200-300 words discussing your most important design decision, one data-quality challenge, how your data cleaning increased trust, one limitation, and one improvement you would make if the dataset were refreshed regularly.

4. Submission Requirements

- Submit your work to your GitHub repository. Include the executed notebooks (.ipynb), ~~source or reproducible download instructions,~~ star-schema diagram, and README.md.
- The notebooks must show outputs and must run from beginning to end using relative paths.
- Do not include python packages, package caches, or unrelated large files.
- In README.md, provide ~~setup instructions,~~ dataset URL, execution order, and any known limitation.
- Clearly acknowledge external code, AI assistance, or adapted examples according to the course academic-integrity policy. You remain responsible for understanding and verifying every submitted result.

5. Assessment Rubric

Component	Marks	Full-credit evidence
Dataset understanding and questions	8	Suitable public dataset; accurate source and grain; concise data dictionary; exactly four meaningful, feasible questions.
Ingestion strategy	12	Correct distinction and justification of full/incremental, push/pull, and batch/streaming; sensible frequency, watermark, recovery, and rerun plan.
Bronze layer	10	Raw data preserved; required tracking metadata; load counts ~~and traceability; no unreported cleaning~~.
Silver layer	20	Relevant cleaning and transformations; correct types; ~~reasoned~~ missing/duplicate handling; validation rules; quarantine/reporting~~; trusted output~~.
Gold star schema	20	Clear fact grain; appropriate fact and dimensions; PK/FK identification; useful measures; ~~referential checks;~~ readable diagram and explanations.
Analytics and communication	20	Four correct Gold-based queries; result tables~~; at least three suitable visuals~~; accurate interpretations tied to the questions.
Reflection	5	Specific discussion of decisions, challenge, checks, limitation, and improvement.
Reproducibility and presentation	5	Runs end to end; logical organization; relative paths; clear Markdown/comments; professional submission and README.md file.
Total	100	

Performance Descriptors

Level	General description
Excellent (85-100%)	Technically correct, reproducible, well justified, and clearly communicated.
Good (70-84%)	Mostly correct and complete, with minor gaps in justification, validation, schema design, or interpretation.
Satisfactory (60-69%)	Core pipeline is present, but some decisions are weakly justified or some outputs/checks are incomplete.
Needs improvement (<60%)	Major layer, query, schema, or deliverable is missing; results are not reproducible; or design choices are not explained.